

EXAM 7 – SPRING 2015

16. (1.5 points)

The discounted cash flow (DCF) method for company valuation relies on the estimation of free cash flows.

a. (0.75 point)

Define free cash flow.

b. (0.25 point)

Briefly explain a practical weakness with the estimation of free cash flows.

c. (0.5 point)

Explain how the abnormal earnings method represents an improvement over the DCF method.

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

QUESTION 16	
TOTAL POINT VALUE: 1.5	LEARNING OBJECTIVE: B2: Value the equity of a P&C insurer based on its expected future dividends, its free cash flow to equity, or its expected abnormal earnings.
SAMPLE ANSWERS	
Part a: 0.75 point	
<i>Sample Answer 1</i> Free cash flow: all cash that could be paid out to the firm's sources of capital, whether or not it is actually paid out in the period it is generated measured net of any amounts required to be reinvested in the firm to maintain operations and generate growth at the rate assumed in the forecasts. <i>Sample Answer 2</i> Free cash flow is the money that can be paid as dividend it's usually net of any cash flow that required to invest for operation and company growth	
Part b: 0.25 point	
<i>Sample Answer 1</i> DCF methods require forecasting and modifying financial statements, the resulting measure may be unfamiliar to management. <i>Sample Answer 2</i> It uses adjusted accounting measures which does not in-line with balance sheet or any financial statement and hence hard to understand/reconcile for management. <i>Sample Answer 3</i> It has a large terminal value. Thus it puts a lot of weight on expected growth rate and discount rate.	
Part c: 0.5 point	
<i>Sample Answer 1</i> Usually DCF method assumes the free cash flow grows in perpetuity, this is unlikely given competitors will enter the market and squeeze the profit. Abnormal earnings method, instead, only assume the abnormal earning exist for a period of time, which is more realistic. <i>Sample Answer 2</i> Abnormal earnings calculates firm value directly using accounting measures, will not need to adjust into a cash flow measure.	
EXAMINERS' REPORT	
This question was challenging in that it was asking for more qualitative details of the Goldfarb paper. Whereas the paper focuses on calculation, this question focused on being able to explain some of the implications of those calculations and formulas. This required either a very strong	

EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

understanding of the nuances of the methods or a very detailed memorization of the paper.
Part a
• Candidates were expected to include at least the portion of the definition of free cash flow that described FCF as money available to pay dividends. • To receive full credit, candidates also had to state that the amount is net of funds required for reinvestment in the company to support normal operations and forecasted growth. • Common errors included giving the definition of the Free Cash Flow on Equity, which was more specific than the general definition we were looking for. Candidates also commonly did not state that the amount was net of amounts needed to support growth and general operations.
Part b
• Candidates were expected to state that due to the many adjustments needed to convert the accounting measures to cash flows, the results may not be familiar to management. • This part was only a quarter point, so there was no partial credit. • The most common incorrect response was to state that a weakness of the method was that you have to project financial values into the future.
Part c
• The abnormal earnings method is an improvement over the discounted cash flow method because it uses values directly from the income statement without adjustment. We also gave credit for explanations that argued that abnormal earnings is an improvement over DCF because DCF assumes growth in perpetuity, whereas abnormal earnings are assumed to converge to 0 over a finite horizon, which is more reasonable. • To get full credit for the second option, the candidate must have explicitly stated that the assumption of the abnormal earnings approach is more reasonable. • Common incorrect answers included stating that abnormal earnings focus on the source of value creation and DCF focuses on the effect of value creation, with no further explanation of why this is an improvement. Many candidates also discussed the reduced weight that the abnormal earnings method puts on the tail value, without explanation of why that was an improvement (some credit was given for this response). • Several candidates approached this part by simply stating what they knew about the two methods. An answer receiving full credit, however, provided a comparison of the methods and demonstrated an understanding of the advantages and disadvantages of the methods.